

Text Summarization Using Hugging Face Transformers

Introduction

Text summarization is a Natural Language Processing (NLP) technique that automatically converts a lengthy document into a shorter version while preserving its key information and meaning. It helps users quickly understand large amounts of text and is widely used in news aggregation, research, education, and business applications. This project demonstrates automatic text summarization using a pre-trained transformer model in Google Colab.

Abstract

The objective of this project was to develop an automatic text summarization system using the Hugging Face Transformers library. Google Colab was used as the development environment, and the facebook/bart-large-cnn pre-trained model was selected for abstractive summarization. The required libraries were installed, and the model was loaded using the Transformers pipeline. The system was tested on both a short paragraph and a longer article related to Artificial Intelligence. The generated summaries successfully captured the main ideas while reducing the overall text length.

Tools Used

Google Colab

Python 3

Hugging Face Transformers (Version 4.46.3)

PyTorch

BART Pre-trained Model (facebook/bart-large-cnn)

Tokenizers

Hugging Face Hub

Steps Involved in Building the Project

Opened Google Colab and installed Transformers version 4.46.3.

Imported the pipeline module from the Transformers library.

Loaded the pre-trained facebook/bart-large-cnn summarization model.

Entered input text and a sample article.

Generated summaries using the summarization pipeline with appropriate parameters (max_length, min_length, and do_sample=False).

Displayed the generated summaries and verified that the important information was preserved.

Conclusion

The project successfully demonstrated the use of transformer-based models for automatic text summarization. The BART model produced accurate and meaningful summaries for both short and long text inputs without requiring model training. Hugging Face Transformers and Google Colab provided an efficient environment for implementing the solution. This project highlights the effectiveness of modern NLP models in reducing reading time while preserving the essential content of documents and can be extended to summarize articles, reports, research papers, and other long-form text.

Platform: Google Colab

Library: Transformers 4.46.3

Model: facebook/bart-large-cnn

Task: Text Summarization using the Hugging Face pipeline API

Testing: Both a short paragraph and a longer AI article